

Q1. Number System

Find the remainder when (3^{45}) is divided by 10.

- a) 1
- b) 3
- c) 7
- d) 9

Q2. HCF–LCM

LCM of two numbers is 420 and HCF is 14. If one number is 70, the other is:

- a) 84
- b) 90
- c) 96
- d) 100

Q3. Simple Interest

A sum becomes ₹7,200 in 4 years at 10% simple interest. Principal is:

- a) 5000
- b) 5200
- c) 5400
- d) 6000

Q4. Compound Interest

₹10,000 becomes ₹12,100 in 2 years at compound interest. Rate is:

- a) 8%
- b) 9%
- c) 10%
- d) 12%

Q5. Percent Change

If the price of sugar increases by 25%, by what % must consumption decrease so that expenditure remains same?

- a) 10%
- b) 15%
- c) 20%
- d) 25%

Q6. Time & Work

A completes a work in 16 days, B in 24 days. They work together for 6 days. Remaining work will be completed by B alone in:

- a) 8 days
- b) 10 days

- c) 12 days
- d) 14 days

Q7. Pipes & Cistern

Pipe A fills a tank in 20 hrs. Pipe B fills it in 30 hrs. Both are opened together but after 6 hrs, pipe B is closed. Remaining tank will be filled by A in:

- a) 6 hrs
- b) 7 hrs
- c) 8 hrs
- d) 9 hrs

Q8. Ratio & Ages

Ratio of ages of A and B is 4:5. After 6 years, ratio becomes 5:6. Present age of A is:

- a) 18
- b) 20
- c) 24
- d) 28

Q9. Average

Average of 12 numbers is 35. If one number 50 is replaced by 26, new average is:

- a) 33
- b) 34
- c) 35
- d) 36

Q10. Train

A train covers 450 m in 30 seconds. Speed is:

- a) 45 km/h
- b) 50 km/h
- c) 54 km/h
- d) 60 km/h

Q11. Boats

A boat's speed in still water is 10 km/h and speed of stream is 2 km/h. Time to travel 24 km upstream is:

- a) 2 hrs
- b) 3 hrs
- c) 4 hrs
- d) 6 hrs

Q12. Permutation

How many 4-digit numbers can be formed using digits 1,2,3,4,5 without repetition?

- a) 60
- b) 120
- c) 240
- d) 360

Q13. Probability

A card is drawn from a deck. Probability that it is a face card is:

- a) $\frac{3}{13}$
- b) $\frac{1}{4}$
- c) $\frac{4}{13}$
- d) $\frac{1}{13}$

Q14. Series

Find the next term:

5, 11, 23, 47, ?

- a) 85
- b) 91
- c) 95
- d) 97

Q15. Profit & Loss

An article is sold at 15% loss. If selling price is ₹850, cost price is:

- a) 900
- b) 950
- c) 1000
- d) 1100

Q16. Mixture

A mixture contains milk and water in ratio 3:2. If 10 liters of water is added, ratio becomes 3:4. Original quantity of mixture is:

- a) 20 L
- b) 25 L
- c) 30 L
- d) 35 L

Q17. Quadratic

If $(x^2 - \underline{5x} + 6 = 0)$, roots are:

- a) 1, 6
- b) 2, 3
- c) -2, -3
- d) 3, 5

Q18. Logarithm

If ($\log_2(x) = 6$), then $x =$

- a) 12
- b) 32
- c) 64
- d) 128

it becomes- 2^6

Q19. Geometry

Sum of angles of a polygon is 1260° . Number of sides is:

- a) 7
- b) 8
- c) 9
- d) 10

Q20. Mensuration

Volume of a cube is 512 cm^3 . Side of cube is:

- a) 6 cm
- b) 7 cm
- c) 8 cm
- d) 9 cm

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Q21. Sentence Correction

Choose correct sentence:

- a) Neither of the answers are correct.
- b) Neither of the answers is correct.
- c) Neither of the answer is correct.
- d) Neither answers are correct.

Q22. Preposition

"He insisted ____ paying the bill."

- a) on
- b) at
- c) for
- d) to

Q23. Para Jumble

Choose correct order:

- A. He reached the station late.
- B. The train had already left.
- C. He missed the train.
- D. He was stuck in traffic.

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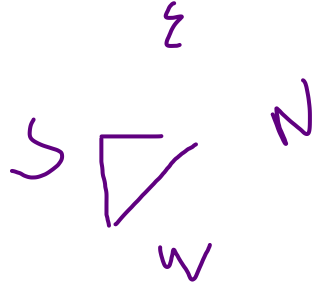
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- a) A D B C
- b) D A B C
- c) D A C B
- d) A B D C

Q24. Direction Sense

A man moves 5 km south, then 12 km west, then 5 km north. Distance from start:

- a) 5 km
- b) 10 km
- c) 12 km
- d) 17 km



Q25. Data Sufficiency

Is $x > y$?

- (1) $x + y = 10$
- (2) $x - y = 4$

- a) Statement (1) alone sufficient
- b) Statement (2) alone sufficient
- c) Both together sufficient
- d) Both together not sufficient

Q26. Java

Which of the following is not a Java feature?

- a) Object Oriented
- b) Platform Independent
- c) Pointer Arithmetic
- d) Robust

Q27. Java Output

`System.out.println(5 + 10 + "Hello" + 5 + 10);`

- a) 15Hello510
- b) 15Hello15
- c) 510Hello510
- d) Error

Q28. OOP

Which concept hides implementation details?

- a) Encapsulation
- b) Abstraction
- c) Polymorphism
- d) Inheritance

Q29. Exception

Which is a checked exception?

- a) ArithmeticException
- b) NullPointerException
- c) IOException
- d) ArrayIndexOutOfBoundsException

unchecked

Q30. Collections

Which collection does not allow duplicates?

- a) ArrayList
- b) LinkedList
- c) HashSet
- d) Vector

Q31. DSA

Which sorting algorithm uses divide and conquer?

- a) Bubble Sort
- b) Merge Sort
- c) Selection Sort
- d) Insertion Sort

Q32. DSA

Which data structure is best for implementing Undo operation?

- a) ~~Queue~~
- b) Stack
- c) Heap
- d) Graph

Q33. OS

Which of the following is a process synchronization tool?

- a) Semaphore
- b) Compiler
- c) Assembler
- d) Loader

Q34. DBMS

Which SQL clause is used to filter groups?

- a) WHERE
- b) GROUP BY
- c) HAVING
- d) ORDER BY

Q35. CN

Which device works at Data Link Layer?

- a) Hub
- ☒ b) Switch
- c) Router
- d) Modem

Q36. Java

Which method is used to start a thread?

- ☒ a) start()
- b) run()
- c) init()
- d) execute()

Q37. DSA

A complete binary tree of height h has maximum nodes:

- a) $2h$
- ☒ b) $(2^{h+1} - 1)$
- c) $(2^h - 1)$
- d) (h^2)

Q38. DBMS

Which command is used to add a new column in an existing table?

- a) UPDATE
- b) INSERT
- ☒ c) ALTER
- d) MODIFY

Q39. Java

Which of the following is an interface in Java?

- a) String
- ☒ b) Runnable
- c) Integer
- d) System

Q40. DSA

In BFS traversal, nodes are visited in:

- a) Depth-first order
- b) Random order
- ☒ c) Level-wise order
- d) Reverse order